LEXICAL SURPRISE SELECTIVELY AMPLIFIES PHONETIC AND SEMANTIC NEURAL CODES: A CODEC EEG STUDY

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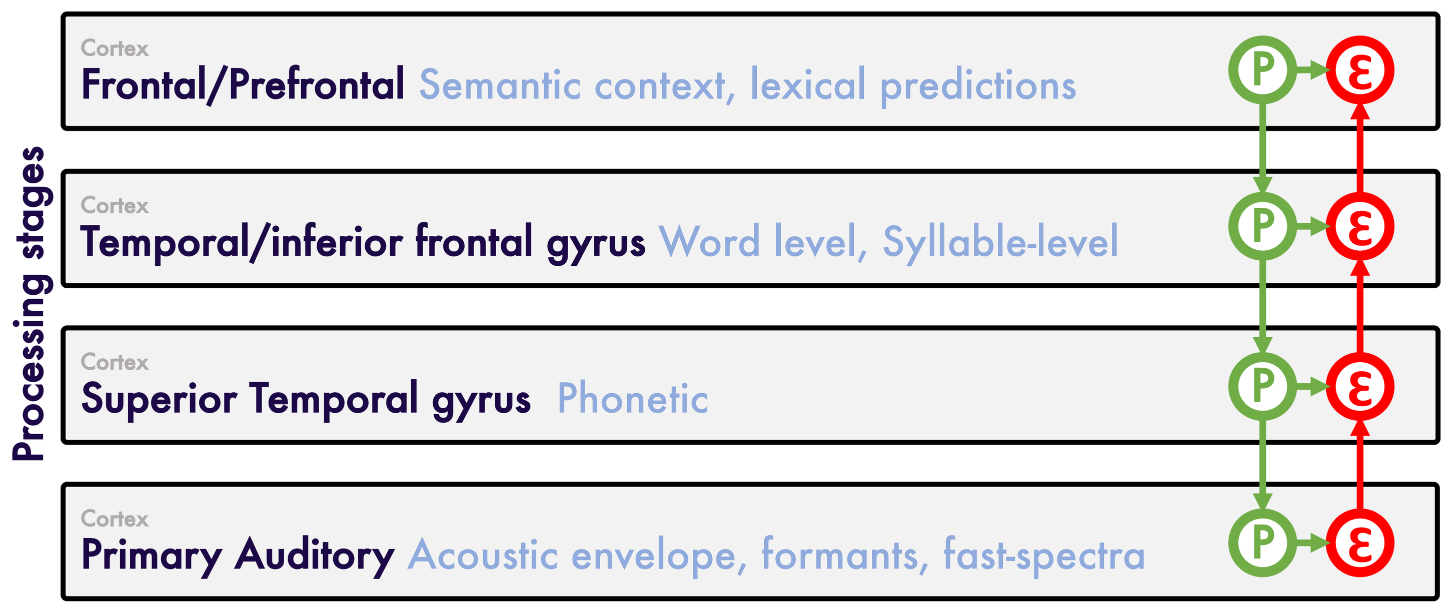
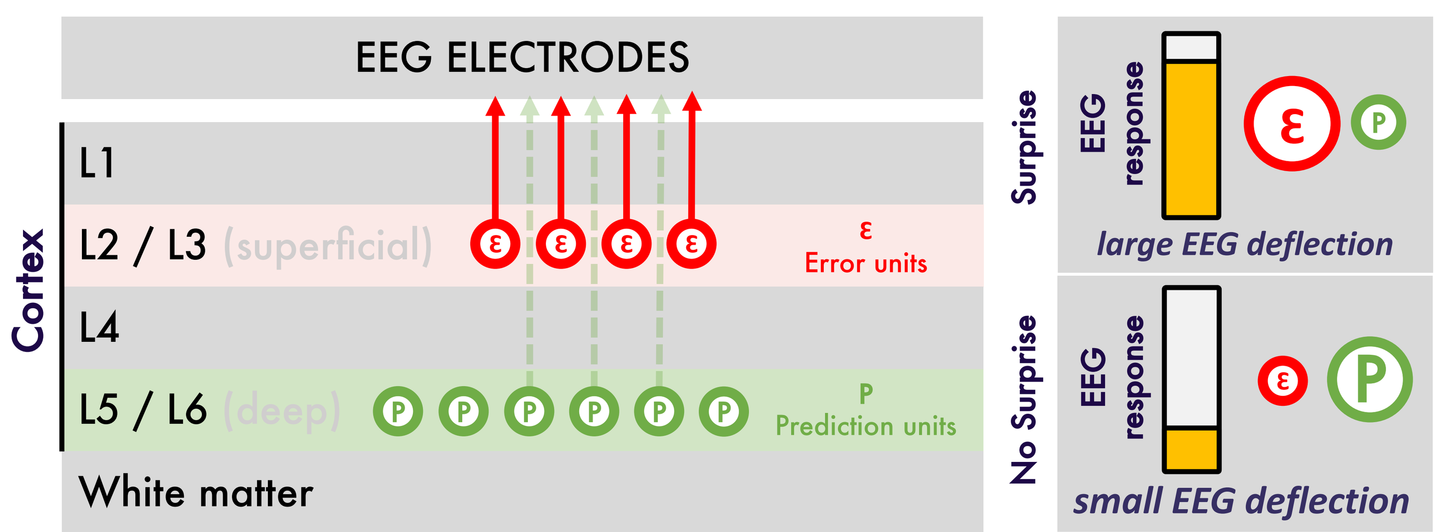
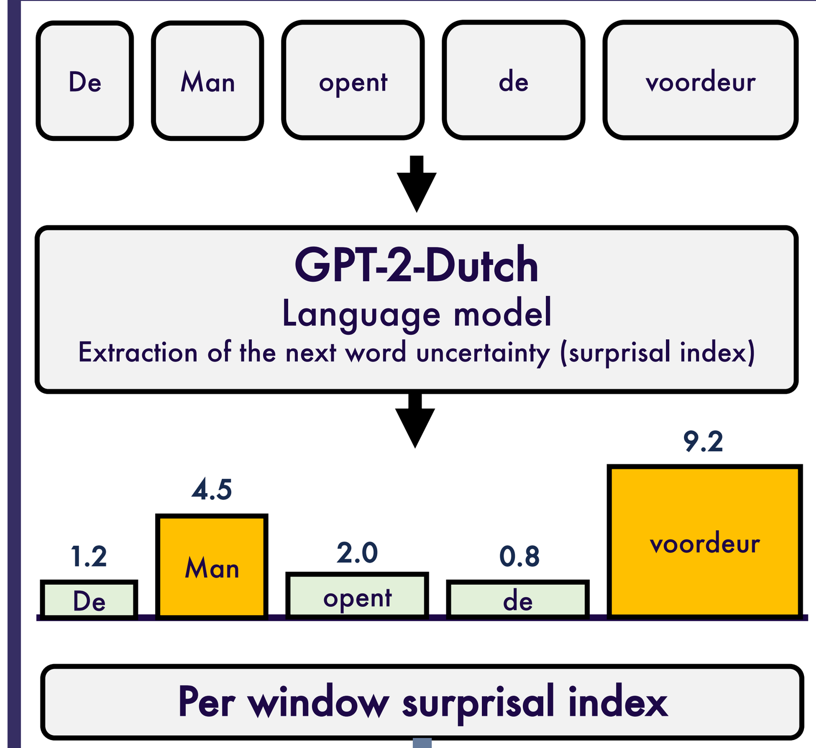
Hierarchical predictive-coding theory *(Friston, 2005; Rao & Ballard, 1999)* predicts that the cortex amplifies the unexpected portion of incoming sensory input - and that this amplification is content-specific: it should appear only at the representational levels carrying the violated prediction. Prior EEG work *(Heilbron 2022; Brodbeck 2018; Gillis 2021)* has used a single speech representation at a time, making content-specificity hard to test cleanly.

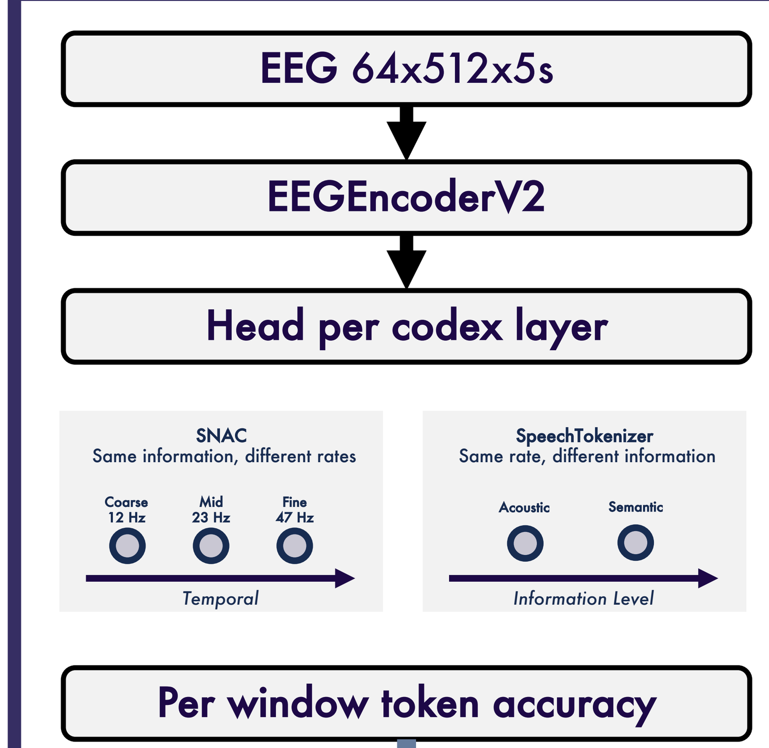




SURPRISAL



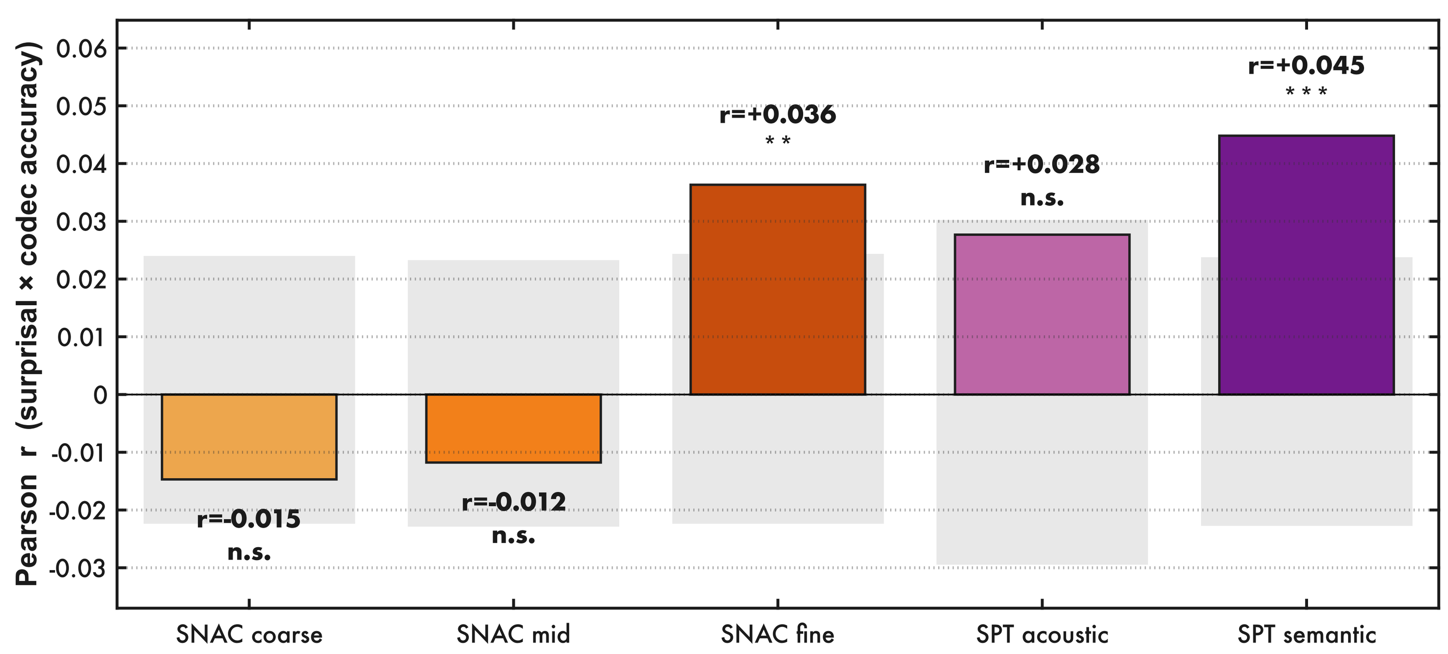
EEG & CODEC



Correlation?

Does the brain selectively amplify content-specific neural codes of speech when its lexical predictions are violated - and if so, which ones, when, and where?

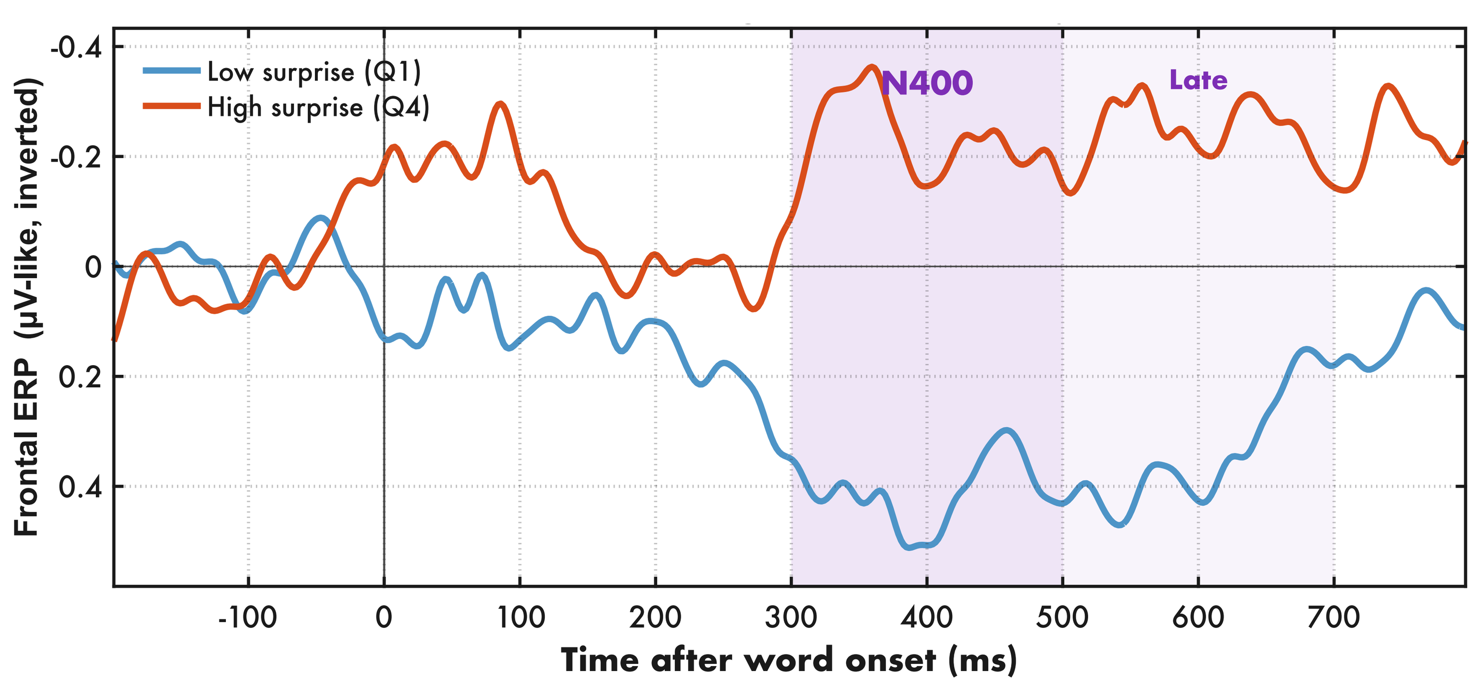
Is it content specific?



Pearson correlation between per-window lexical surprisal (GPT-2-Dutch, bits) and per-window EEG-decoding accuracy, computed separately for each of the five codec streams. Grey bands behind each bar show the permutation-null 95% confidence interval (10,000 surprisal-label shuffles). Only the high-rate phonemic stream (SNAC fine, $r = +0.036$, $p = .003$) and the lexical-semantic stream (spt_semantic, $r = +0.045$, $p < .001$) lie outside the null distribution; the envelope (SNAC coarse), syllabic (SNAC mid) and acoustic-timbre (SPT acoustic) streams are inside the null. The dissociation across architecturally-matched decoder heads – SNAC coarse and SNAC fine share both head architecture and vocabulary - rules out an architectural or generic-arousal explanation.

Pearson r between lexical surprisal and EEG decoding accuracy is positive at the phonemic (SNAC fine, $r = +0.036$, $p = .003$) and semantic (SPT Semantic, $r = +0.045$, $p < .001$) codec streams and null at the envelope, syllabic and timbre streams.

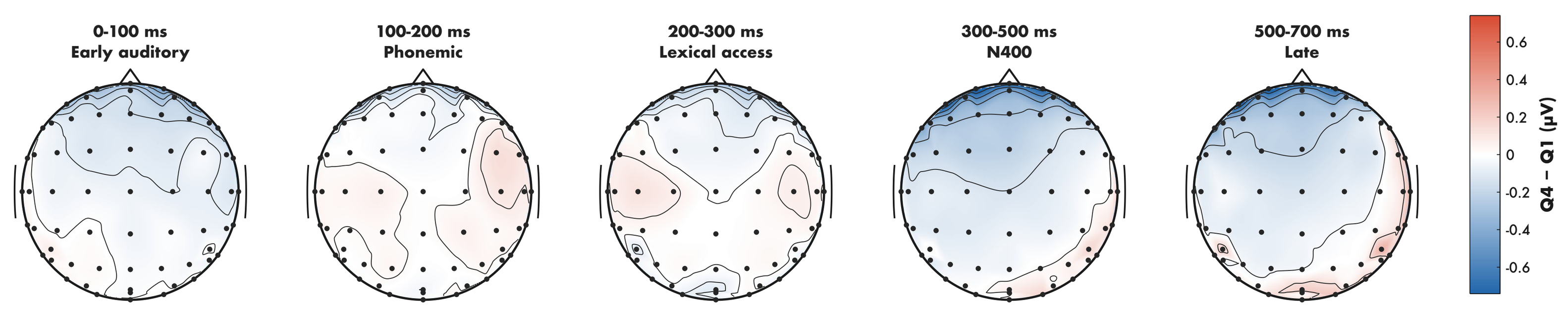
When does it happen?



Grand-average ERP at frontal channels (Fp1/Fpz/Fp2/AF7/AF3/AFz/AF4/AF8) for low- vs high-surprise words (Q1 vs Q4 of the GPT-2 surprisal distribution; 14-subject cohort, y-axis inverted so negative-going deflections point upward). High-surprise words (orange) diverge from low-surprise words (blue) at 300 ms and remain more negative throughout the classical N400 window (300-500 ms, purple band) and the late re-analysis window (500-700 ms). The shape and timing match the canonical N400 elicited by contextually unexpected words during natural speech listening.

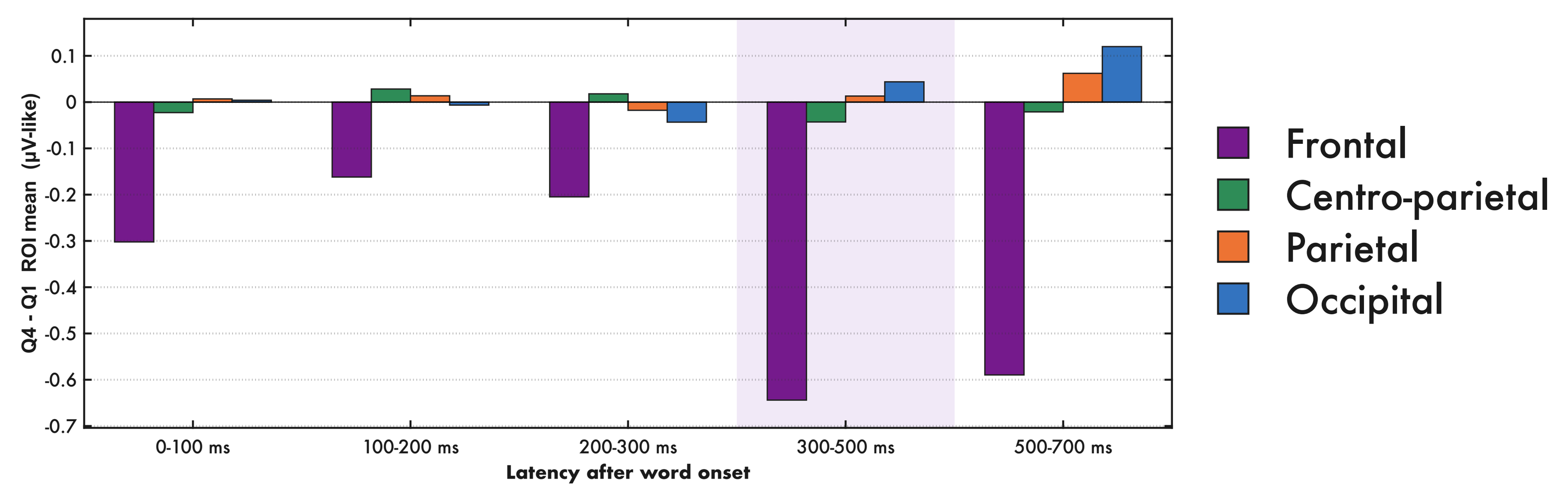
Grand-average frontal-channel ERP for low- vs high-surprise words: a clear N400-stage divergence at 300-500 ms post-onset.

Where in the cortex?



Bilinear-interpolated Q4 – Q1 topomap row across the five latency. The early-auditory, phonemic, and lexical-access bins are nearly flat; a frontal-anterior negativity emerges sharply at the N400 bin (300-500 ms) and persists with similar topography in the late bin (500-700 ms). The wavefront-like progression visualises the integration-stage timing seen in the regional-bar quantification.

Five Q4 – Q1 scalp topomaps from 0-100 ms through 500-700 ms. The frontal negativity emerges at the N400 column and persists into the late window.



Mean Q4 – Q1 ERP amplitude (high-surprise minus low-surprise word events) within four bilateral scalp ROIs across five post-onset latency bins. The frontal ROI (purple) shows a small early negativity and a much larger negativity at the N400 window, persisting into the late re-analysis window; centro-parietal, parietal and occipital ROIs remain near zero across all bins. The 2× ratio of frontal-ROI N400 amplitude to early-auditory amplitude indicates an integration-stage origin rather than bottom-up sensory amplification.

The Q4 – Q1 frontal-ROI effect peaks at the N400 window (-0.64 , 4× the early-auditory bin) and persists into the late re-analysis window; parietal/occipital ROIs stay at zero.

CONCLUSION

Does the brain selectively amplify content-specific neural codes of speech when its lexical predictions are violated - and if so, which ones, when, and where?

Which?	When?	Where?
phonemic + semantic only	N400 integration stage (not sensory)	Left Inferior Frontal Girus (Broca's area)